

## 4 Bearings, Bushings, and Other Shit That Spins

### 4.1 Boundary-Lubricated Bearings (Bushings)

#### 4.1.1 Anatomy

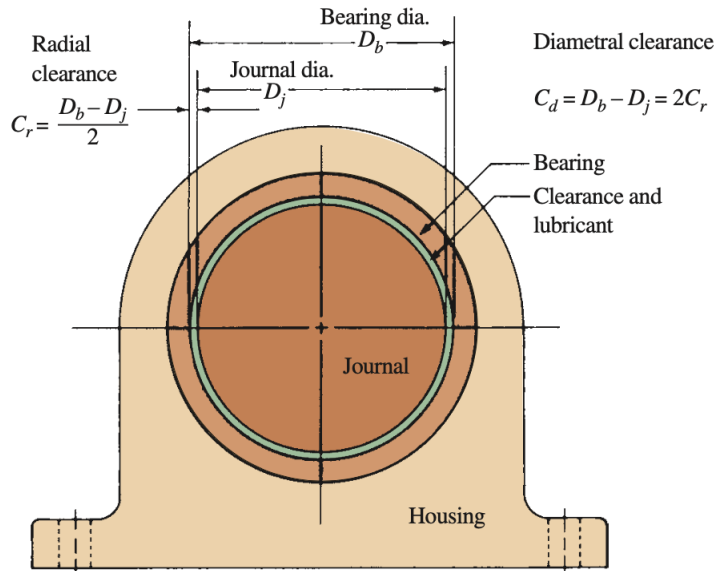


FIGURE 16-1 Bearing geometry

#### 4.1.2 Design Selection

1. Get random factors from these tables based on your bearing type/material,

**Table 12–9 Wear Factors in U.S. Customary Units\***

| Bushing Material         | Wear Factor $K$ | Limiting $PV$ |
|--------------------------|-----------------|---------------|
| Oiles 800                | $3(10^{-10})$   | 18 000        |
| Oiles 500                | $0.6(10^{-10})$ | 46 700        |
| Polyactal copolymer      | $50(10^{-10})$  | 5 000         |
| Polyactal homopolymer    | $60(10^{-10})$  | 3 000         |
| 66 nylon                 | $200(10^{-10})$ | 2 000         |
| 66 nylon + 15% PTFE      | $13(10^{-10})$  | 7 000         |
| + 15% PTFE + 30% glass   | $16(10^{-10})$  | 10 000        |
| + 2.5% MoS <sub>2</sub>  | $200(10^{-10})$ | 2 000         |
| 6 nylon                  | $200(10^{-10})$ | 2 000         |
| Polycarbonate + 15% PTFE | $75(10^{-10})$  | 7 000         |
| Sintered bronze          | $102(10^{-10})$ | 8 500         |
| Phenol + 25% glass fiber | $8(10^{-10})$   | 11 500        |

\*dim[ $K$ ] = in<sup>3</sup> · min/(lbf · ft · h), dim [PV] = psi · ft/min.

Source: Data from Oiles America Corp., Plymouth, MI 48170.

**Table 12–10 Coefficients of Friction**

| Type      | Bearing     | $f_s$ |
|-----------|-------------|-------|
| Placetic  | Oiles 80    | 0.05  |
| Composite | Drymet ST   | 0.03  |
|           | Toughmet    | 0.05  |
| Met       | Cermet M    | 0.05  |
|           | Oiles 2000  | 0.03  |
|           | Oiles 300   | 0.03  |
|           | Oiles 500SP | 0.03  |

2. Calculate minimum bearing length  $L$  using the following equation:

$$L \geq \frac{720 f_s n_d F N}{J \bar{h}_{CR} (T_f - T_\infty)}$$

where  $f_s$  is coefficient of friction,

$n_d$  is the design factor,

$F$  is the radial load,

$N$  is the angular speed of the bearing,

$T_f$  is the lubricant temperature/max temperature

$T_\infty$  is the ambient temperature

For some reason, we just randomly use  $\bar{h}_{CR} = 2.7$  and  $J = 778$  unless some other values are

given.

Also, use  $T_{\infty} = 70$  if no value is given cause that's what they use in the textbook.

3. Choose a bushing from this table that exceeds the minimum length. Make sure the inner diameter (ID) is bigger than the shaft diameter if that is given.

**Table 12–12 Available Bushing Sizes (in inches) of One Manufacturer\***

|                |                | $L$           |               |               |               |   |                |                |                |   |                |   |                |   |   |
|----------------|----------------|---------------|---------------|---------------|---------------|---|----------------|----------------|----------------|---|----------------|---|----------------|---|---|
| ID             | OD             | $\frac{1}{2}$ | $\frac{5}{8}$ | $\frac{3}{4}$ | $\frac{7}{8}$ | 1 | $1\frac{1}{4}$ | $1\frac{1}{2}$ | $1\frac{3}{4}$ | 2 | $2\frac{1}{2}$ | 3 | $3\frac{1}{2}$ | 4 | 5 |
| $\frac{1}{2}$  | $\frac{3}{4}$  | •             | •             | •             | •             | • |                |                |                |   |                |   |                |   |   |
| $\frac{5}{8}$  | $\frac{7}{8}$  |               | •             | •             |               | • |                | •              |                |   |                |   |                |   |   |
| $\frac{3}{4}$  | $1\frac{1}{8}$ |               | •             | •             |               | • |                | •              |                |   |                |   |                |   |   |
| $\frac{7}{8}$  | $1\frac{1}{4}$ |               |               | •             |               | • | •              | •              |                |   |                |   |                |   |   |
| 1              | $1\frac{3}{8}$ |               |               | •             |               | • | •              | •              | •              | • |                |   |                |   |   |
| 1              | $1\frac{1}{2}$ |               |               | •             |               | • |                | •              |                | • |                |   |                |   |   |
| $1\frac{1}{4}$ | $1\frac{5}{8}$ |               |               |               |               | • | •              | •              | •              | • |                |   |                |   |   |
| $1\frac{1}{2}$ | 2              |               |               |               |               | • | •              | •              | •              | • |                |   |                |   |   |
| $1\frac{3}{4}$ | $2\frac{1}{4}$ |               |               |               |               |   | •              | •              | •              | • | •              | • | •              | • |   |
| 2              | $2\frac{1}{2}$ |               |               |               |               |   |                | •              |                | • | •              | • |                |   |   |
| $2\frac{1}{4}$ | $2\frac{3}{4}$ |               |               |               |               |   |                | •              |                | • | •              | • |                |   |   |
| $2\frac{1}{2}$ | 3              |               |               |               |               |   |                | •              |                | • |                | • |                |   |   |
| $2\frac{3}{4}$ | $3\frac{3}{8}$ |               |               |               |               |   |                | •              |                | • | •              | • |                |   |   |
| 3              | $3\frac{5}{8}$ |               |               |               |               |   |                |                |                | • | •              | • | •              |   |   |
| $3\frac{1}{2}$ | $4\frac{1}{8}$ |               |               |               |               |   |                |                |                | • |                | • |                | • |   |
| 4              | $4\frac{3}{4}$ |               |               |               |               |   |                |                |                | • |                | • |                | • |   |
| $4\frac{1}{2}$ | $5\frac{3}{8}$ |               |               |               |               |   |                |                |                |   |                | • |                | • | • |
| 5              | 6              |               |               |               |               |   |                |                |                |   |                | • |                | • | • |

\*In a display such as this a manufacturer is likely to show catalog numbers where the • appears.

4. Make sure the bushing's length-to-inner diameter (ID) ratio falls within this range:

$$0.5 \geq L/D \geq 2$$

If it doesn't, choose another bushing that does.

5. To make sure your bearing is satisfactory, we will calculate random values and make sure this fall within the acceptable ranges (assuming your bushing is an Oiles 500 SP type):

**Table 12–11 Oiles 500 SP (SPBN • SPWN) Service Range and Properties**

| Service Range                      | Units              | Allowable |
|------------------------------------|--------------------|-----------|
| Characteristic pressure $P_{\max}$ | psi                | <3560     |
| Velocity $V_{\max}$                | ft/min             | <100      |
| PV product                         | (psi)(ft/min)      | <46 700   |
| Temperature $T$                    | °F                 | <300      |
| Properties                         | Test Method, Units | Value     |
| Tensile strength                   | (ASTM E8) psi      | >110 000  |
| Elongation                         | (ASTM E8) %        | >12       |
| Compressive strength               | (ASTM E9) psi      | 49 770    |
| Brinell hardness                   | (ASTM E10) HB      | >210      |
| Coefficient of thermal expansion   | ( $10^{-5}$ )°C    | >1.6      |
| Specific gravity                   |                    | 8.2       |

6. Calculate the characteristic pressure (in psi) and make sure it's in the right range:

$$P_{\max} = \frac{4 n_d F}{\pi DL}$$

where  $F$  is the radial load in lbf,

$n_d$  is the angular speed of the bearing in rpm,

$D = ID$  is the inner bearing diameter in inches,

$L$  is the length of the bearing in inches

7. Calculate the nominal pressure (in psi):

$$P = \frac{n_d F}{DL}$$

8. Calculate the velocity (in ft/min) and make sure it is less than  $V_{\max}$ :

$$V = \frac{\pi DN}{12}$$

9. Calculate  $PV$  by multiplying  $P$  and  $V$ . Make sure it's in the acceptable range.

10. If a maximum wear value was given in the question, calculate the linear wear  $w$ .

$$w = \frac{K n_d F N t}{3L}$$

where  $K$  is the wear factor,

$t$  is the time in hours

11. If all values are within acceptable ranges, congrats you have selected an acceptable bushing!